

Fake News Detection

Logistic Regression, SVM, XGBoost Classifier

Reproduced From: (Jain et al., 2025)

<https://doi.org/10.1007/s13278-025-01455-6>

Submitted by

Fahad Thakur

11997575

Content



Introduction

Data Used

Research Article's Solution

Preprocessing Steps

Model Implementation

Optimization & CV

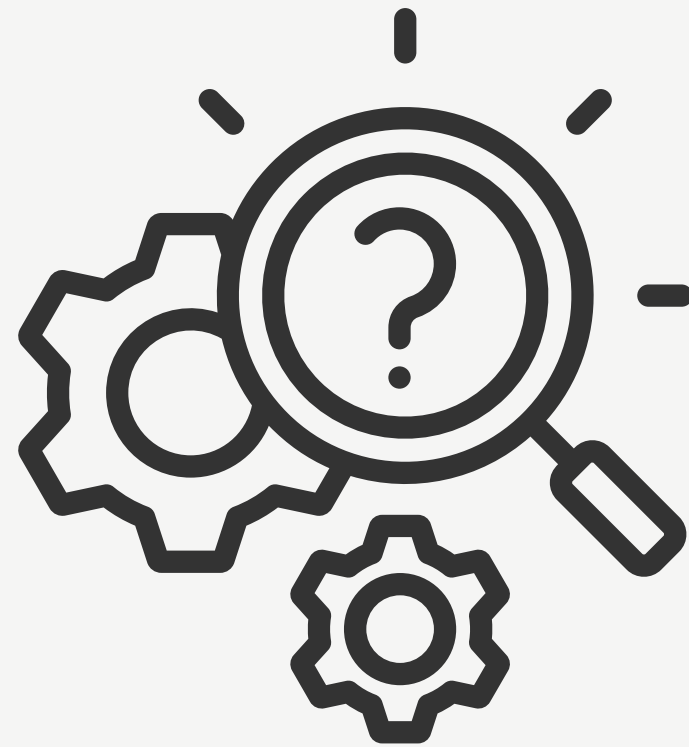
Evaluation Techniques

Model Results

Discussions

GenAI Implementation

Introduction to the Selected Research Article



Fake news spreads through online forums, social media, and can have potential to harm lives and poses a significant threat to society. This can include propaganda, clickbait, conspiracies and more.



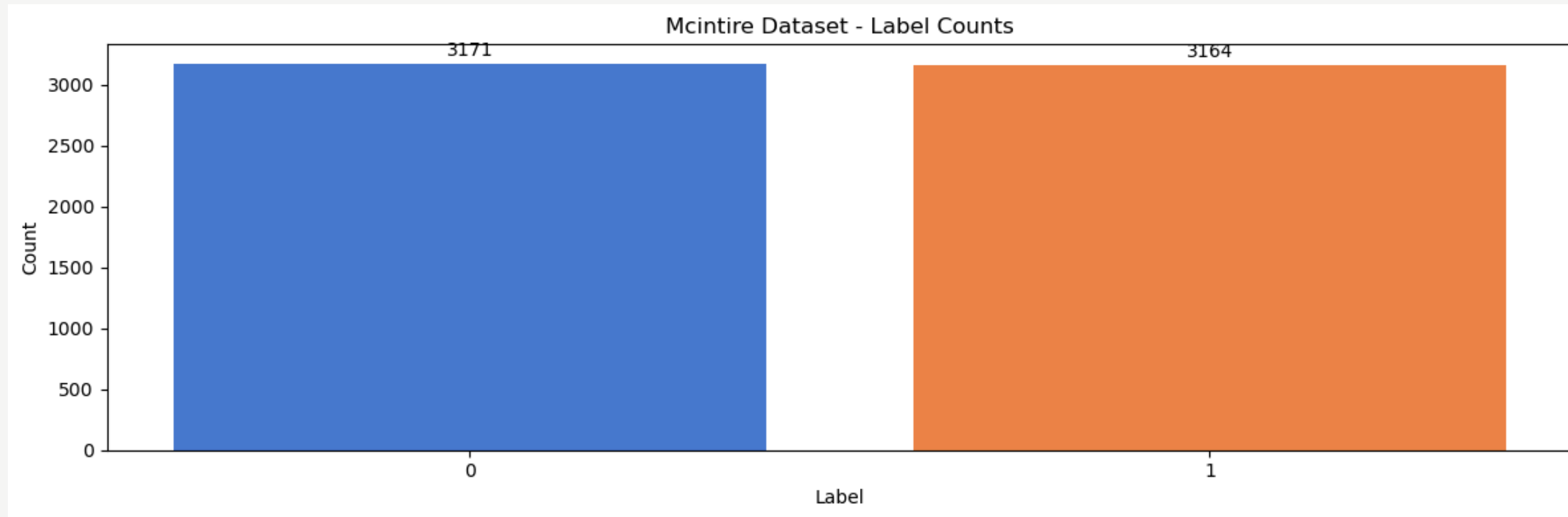
To overcome feature redundancy, suboptimal feature selection limitations and build a more accurate and generalizable model.



The authors implement a meta-heuristic Harris Hawks Optimization (HHO) algorithm as a feature selector to a CNN-BiLSTM hybrid model.

DATA USED

MCIntire Dataset



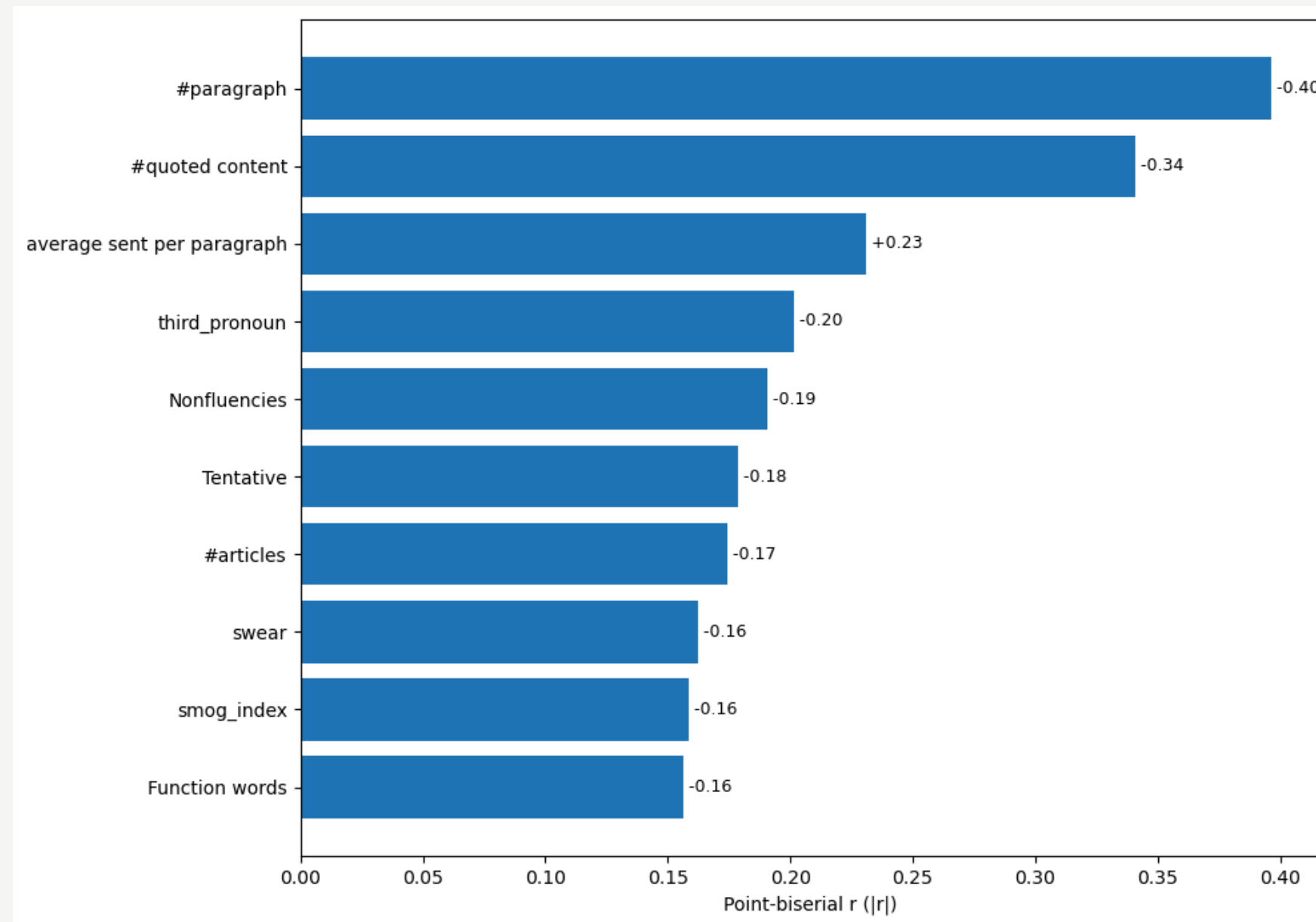
Description

The dataset consists of 6,335 records of data, primarily focusing on both genuine and fake articles of the 2016 US elections.

Dataset Labels

The dataset consists of 3,171 genuine articles, and 3,164 fake articles. Consists of an even distribution of classes. This allows the models to learn patterns in the data without being biased towards a specific class.

DATA USED



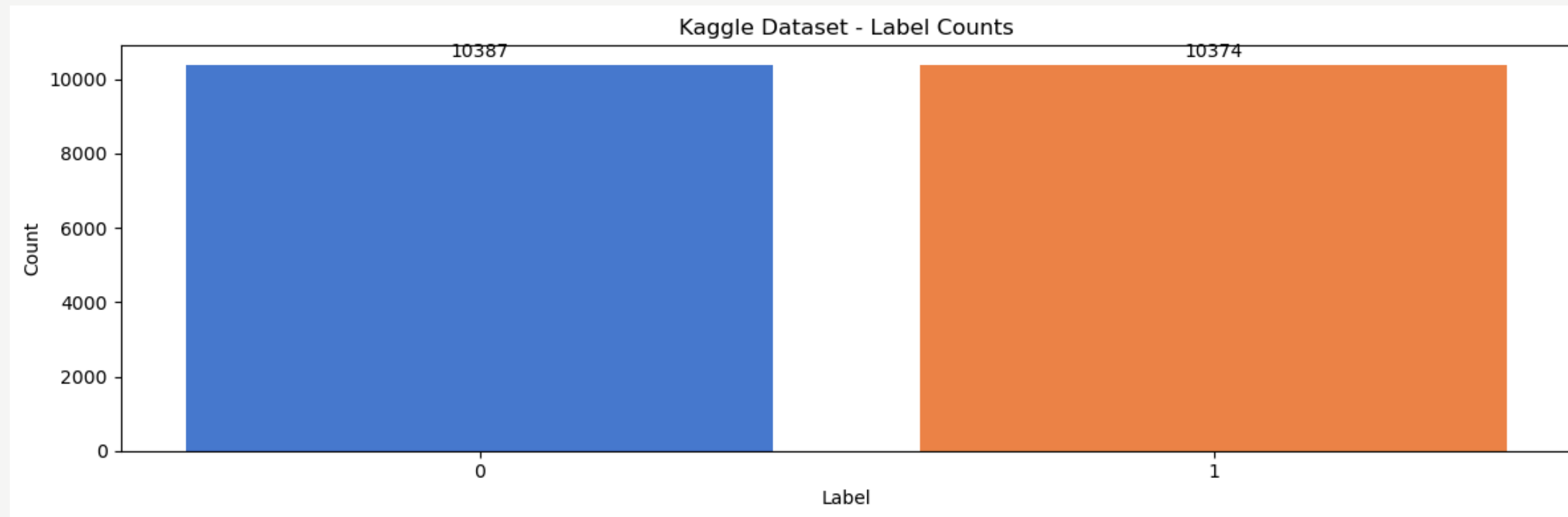
MCIntire Dataset Linguistic Features

The number of paragraphs have a negative correlation with whether the input is a fake news or not.

Having a higher number of sentences per paragraph has a higher correlation with it being fake news.

DATA USED

Kaggle Dataset



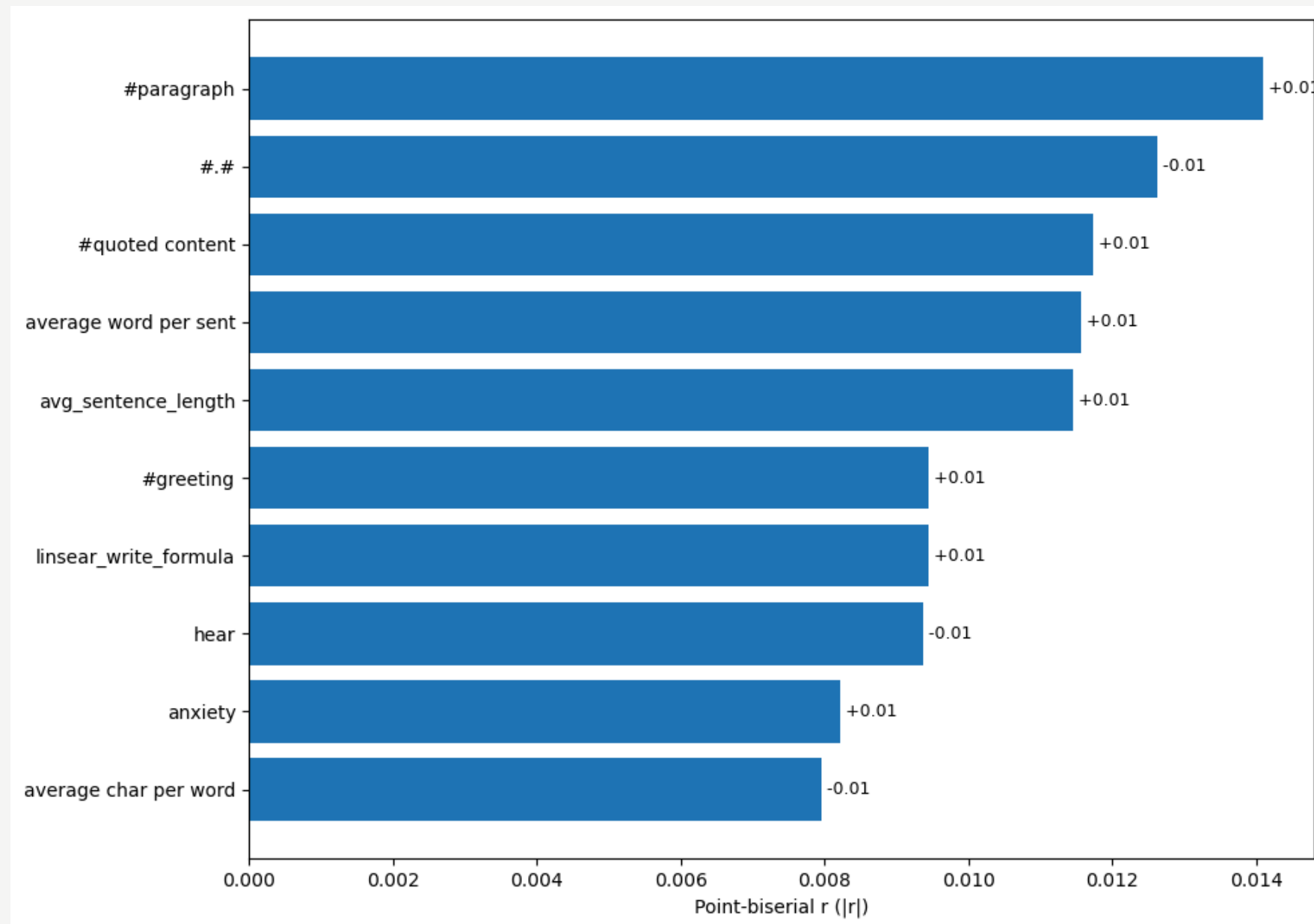
Description

The dataset consists of 20,761 records of data, from various real-world outlets in different domains.

Dataset Labels

The dataset consists of 10,387 genuine articles, and 10,374 fake articles. Consists of an even distribution of classes as well.

DATA USED



Kaggle Dataset Linguistic Features

There is a much lesser correlation between the linguistic features in the Kaggle Dataset with it being fake or not.

Research Article's Solution

$$\partial(t+1) = \begin{cases} \partial_{rand}(t) - p_1 \|\partial_{rand}(t) - 2p_2 \partial(t)\| & \text{if } d \geq 0.5 \\ (\partial_{rabbit}(t) - \partial_m(t)) - p_3(LB + p_4(UB - LB)) & \text{if } d < 0.5 \end{cases} \quad (1)$$

$$\partial_m(t) = \frac{1}{N} \sum_{i=1}^N \partial_i(t) \quad (2)$$

HHO Algorithm

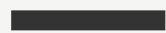
- Preprocess the datasets, then extract its linguistic features. Use Word2Vec to extract embeddings.
- Implement the HHO algorithm to pick the most informative subset of these linguistic features.
- Combine the picked features with the word embeddings.
- Implement a CNN-BiLSTM hybrid model, train and test individually on the 4 datasets used.
- Eq. (1) — Diversification move: At each step, a hawk either (a) orbits a random peer when $d \geq 0.5$, or (b) heads toward the best (“rabbit”) relative to the swarm mean with a small bounded jitter $LB + p_4(UB - LB)$ when $d < 0.5$. This balances broad scouting vs. guided search.

Preprocessing Steps

Adapted from the research article

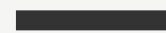
1

Normalization and
convert to lowercase



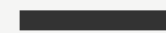
2

Removing URLs and
special characters



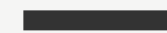
3

Removing stop words
and lemmatize



4

Tokenize using NLTK
tokenizer



Model Implementation

Approach 1

Word2Vec technique adapted from the research article. It converts words into numerical vectors, by placing words with similar meanings in close proximity.
Using the inputs used in the research article:
vector_size = 100, window = 5, min_count = 1

Approach 2

TFIDF Vectorizer to convert text into a matrix of TF-IDF features. It measures the importance of a word in a document based on its frequency and its frequency in the entire corpus.
max_features = 5000

These vectors are then combined with the linguistic features of the dataset, and is then passed to the models to train and evaluate.

Model Implementation

Logistic Regression

Linear model that uses sigmoid function. Can implement L1/L2 regularization to reduce the effect of correlated linguistic features.

SVM

Linear SVMs can excel on TF-IDF features and embeddings. It maximizes the margin between the classes, and is robust to irrelevant tokens.

XGBoost Classifier

Gradient boosted trees, captures non-linear patterns and feature interactions effectively through embeddings that can be passed.

Optimization & Cross Validation

GridSearchCV has been used to search for the most optimal hyper-parameters

Logistic Regression

Hyperparameters
considered:
C: [0.01, 0.1, 1, 10, 100],
penalty: ['l1', 'l2']

SVM

Hyperparameters
considered:
C: [0.1, 1, 10],
kernel: ['linear', 'rbf'],
gamma: ['scale', 'auto']

XGBoost Classifier

Hyperparameters
considered:
n_estimators: [50, 100, 200],
max_depth: [3, 5, 7],
learning_rate: [0.01, 0.1,
0.2],
subsample: [0.8, 1.0]

Optimization & Cross-Validation

Grid Search Hyper-parameters Selected - Logistic Regression

Dataset & Approach	C	Penalty
McIntire W2V Model	1	L2
McIntire TF-IDF Model	100	L2
Kaggle W2V Model	100	L2
Kaggle TF-IDF Model	100	L2

Optimization & Cross-Validation

Grid Search Hyper-parameters Selected - SVM

Dataset & Approach	C	Kernel	Gamma
McIntire W2V Model	10	RBF	Auto
McIntire TF-IDF Model	1	Linear	Scale
Kaggle W2V Model	10	Linear	Scale
Kaggle TF-IDF Model	10	Linear	Scale

Optimization & Cross-Validation

Grid Search Hyper-parameters Selected - XGBoost Classifier

Dataset & Approach	n_estimators	max_depth	learning_rate	subsample
McIntire W2V Model	200	7	0.2	1.0
McIntire TF-IDF Model	200	5	0.2	0.8
Kaggle W2V Model	200	3	0.2	1.0
Kaggle TF-IDF Model	200	3	0.2	0.8

Evaluation Techniques

1

Accuracy

$(TP + TN) / (TP + TN + FP + FN)$

—

2

AUC Score

Higher is better

—

3

Confusion Matrix

Summarize where the model
performed better or
underperformed

—

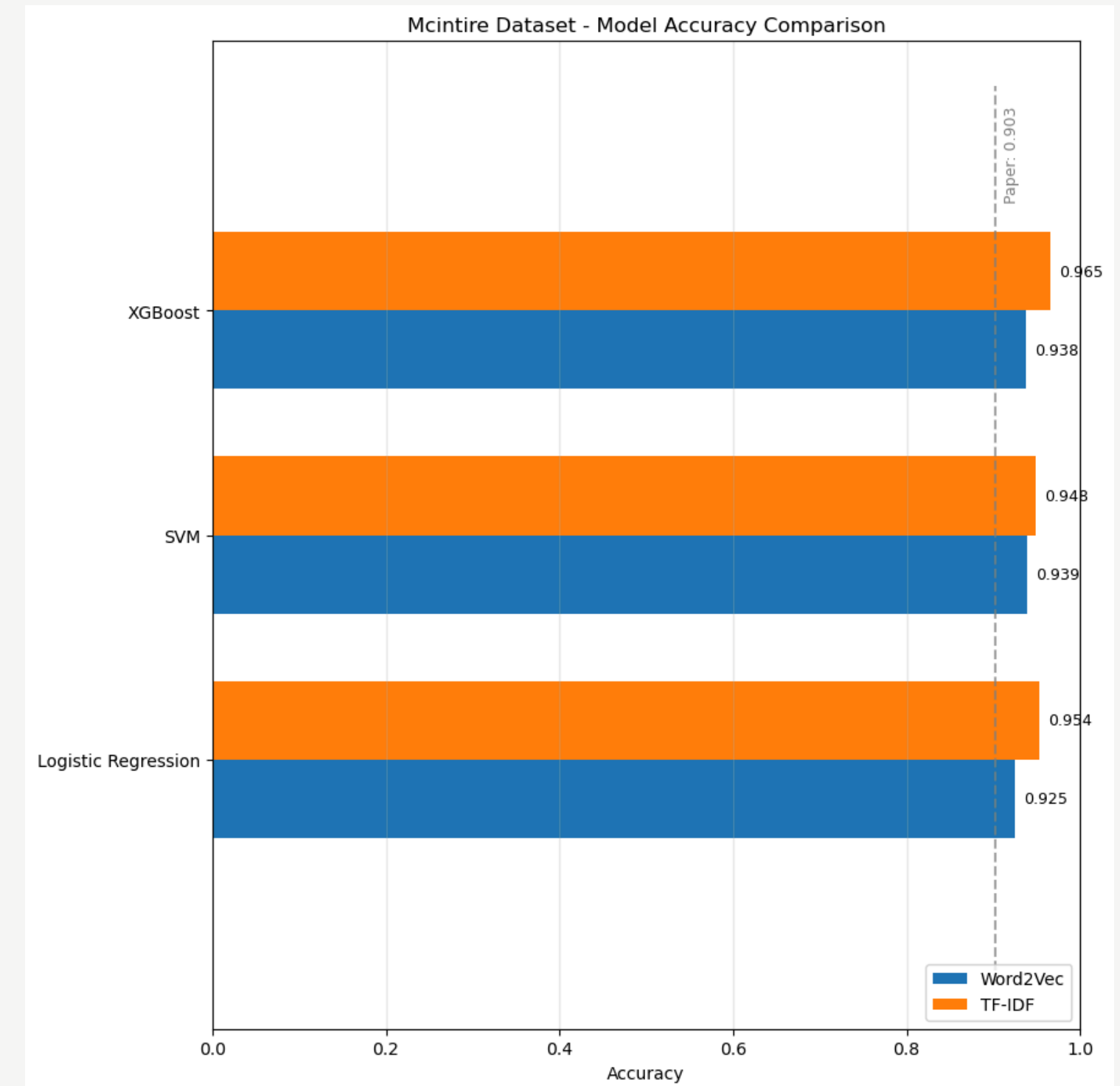
4

OOD Generalization
Check

—

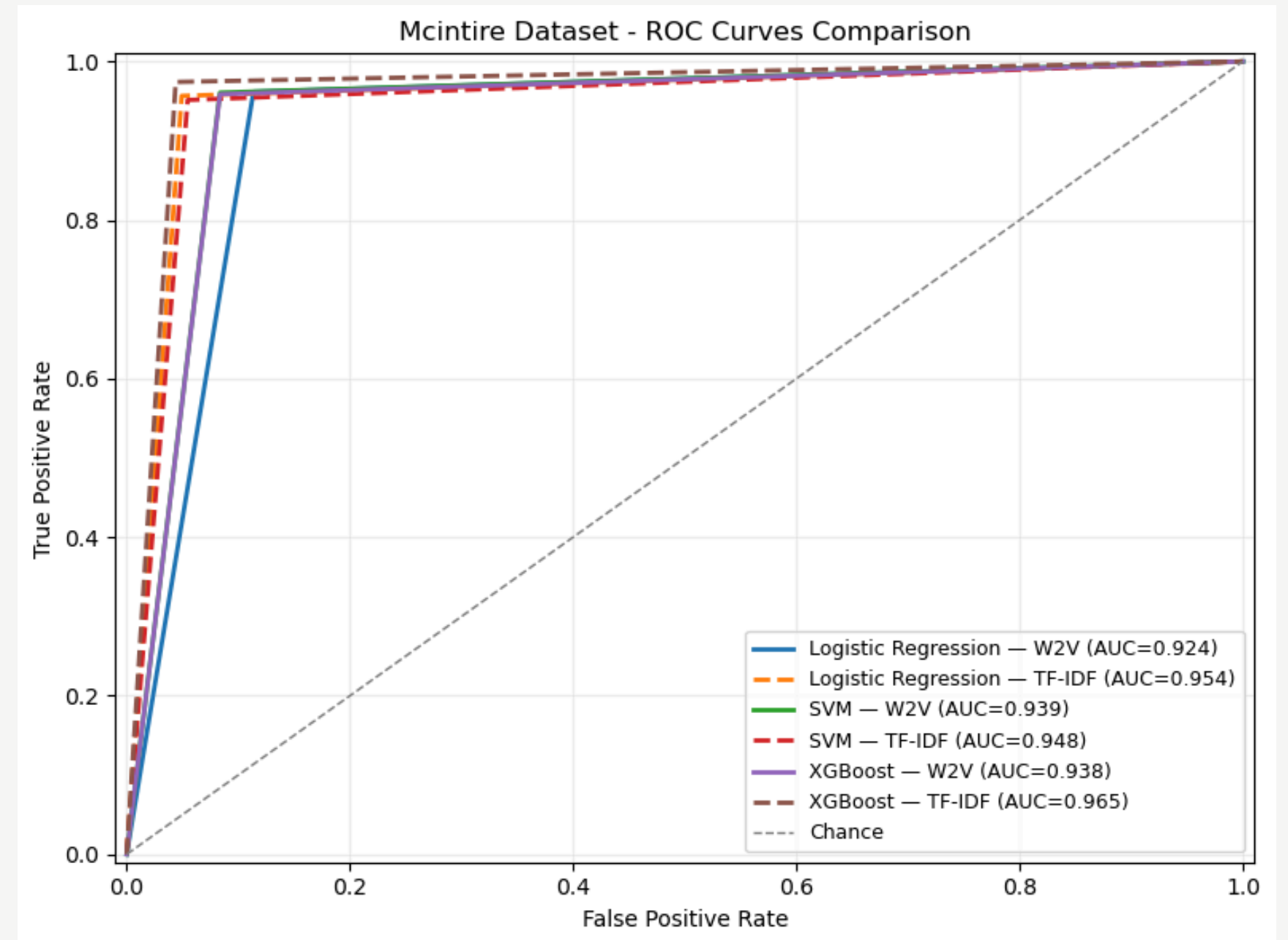
Results - McIntire Dataset

- The paper's accuracy for this dataset was 90.26%.
- All 3 methods that I have used outperform the paper's accuracy by a significant margin.
- The XGBoost model with the TF-IDF approach outperforms all the other approaches with an accuracy of 96.5%.
- Word2Vec approach used by paper shows a lower accuracy in comparison.



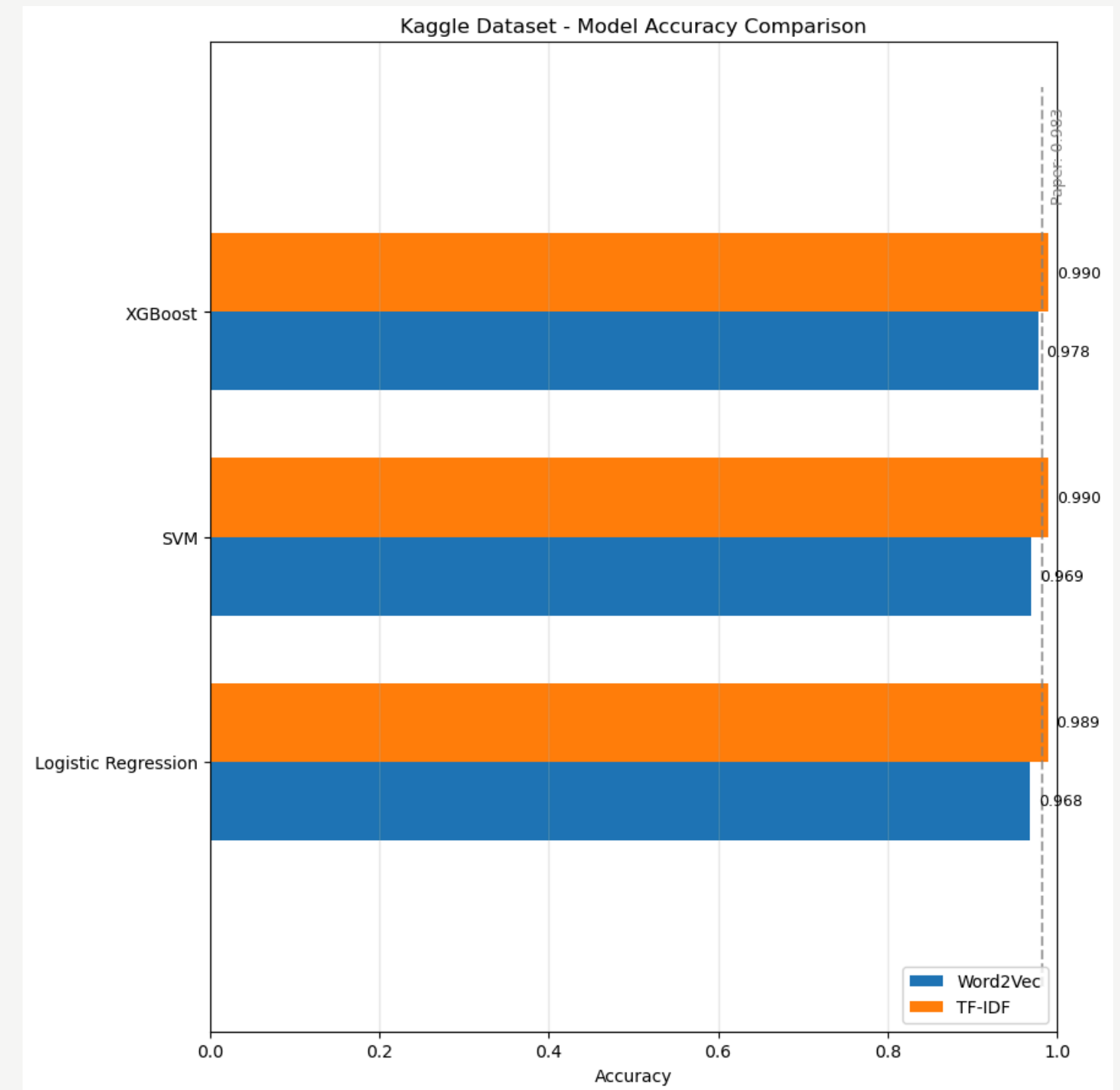
Results - McIntire Dataset

- The AUC curves comparison confirms that the XGBoost model with the TF-IDF approach performs the best with an AUC score of 0.965.
- This is closely followed by the logistic regression model using the TF-IDF approach (AUC = 0.954).
- TF-IDF approaches perform better because it preserves the specific phrase markers and has sparse cues that XGBoost can isolate better.



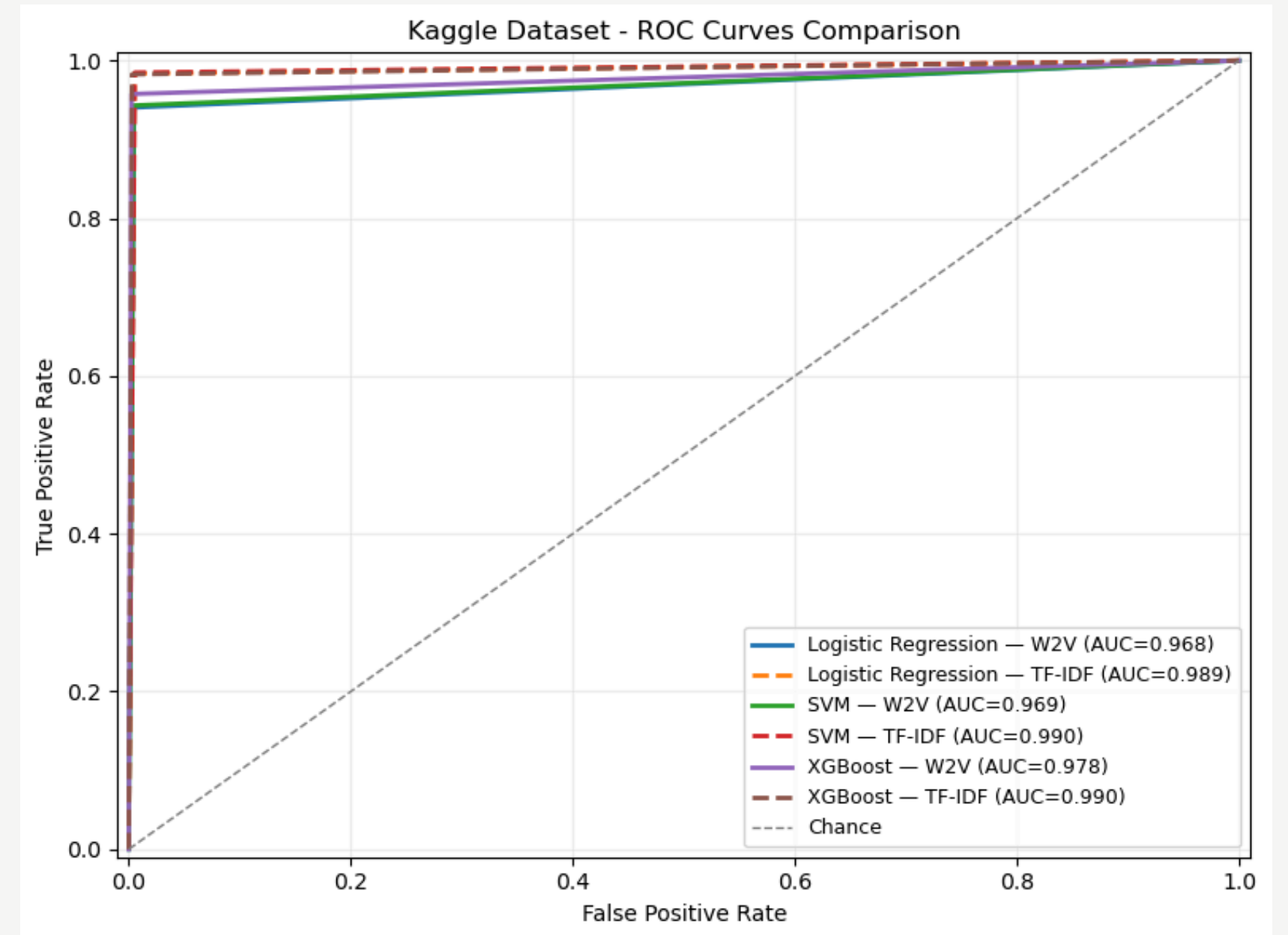
Results - Kaggle Dataset

- The paper's accuracy for this dataset was 98.25%.
- All 3 methods used performed relatively similar to the paper in terms of accuracy.
- The XGBoost and SVM models with the TF-IDF approach outperforms all the other approaches with an accuracy of 99%.
- Once again we see Word2Vec approach used by paper shows a lower accuracy in comparison to the TF-IDF approach.



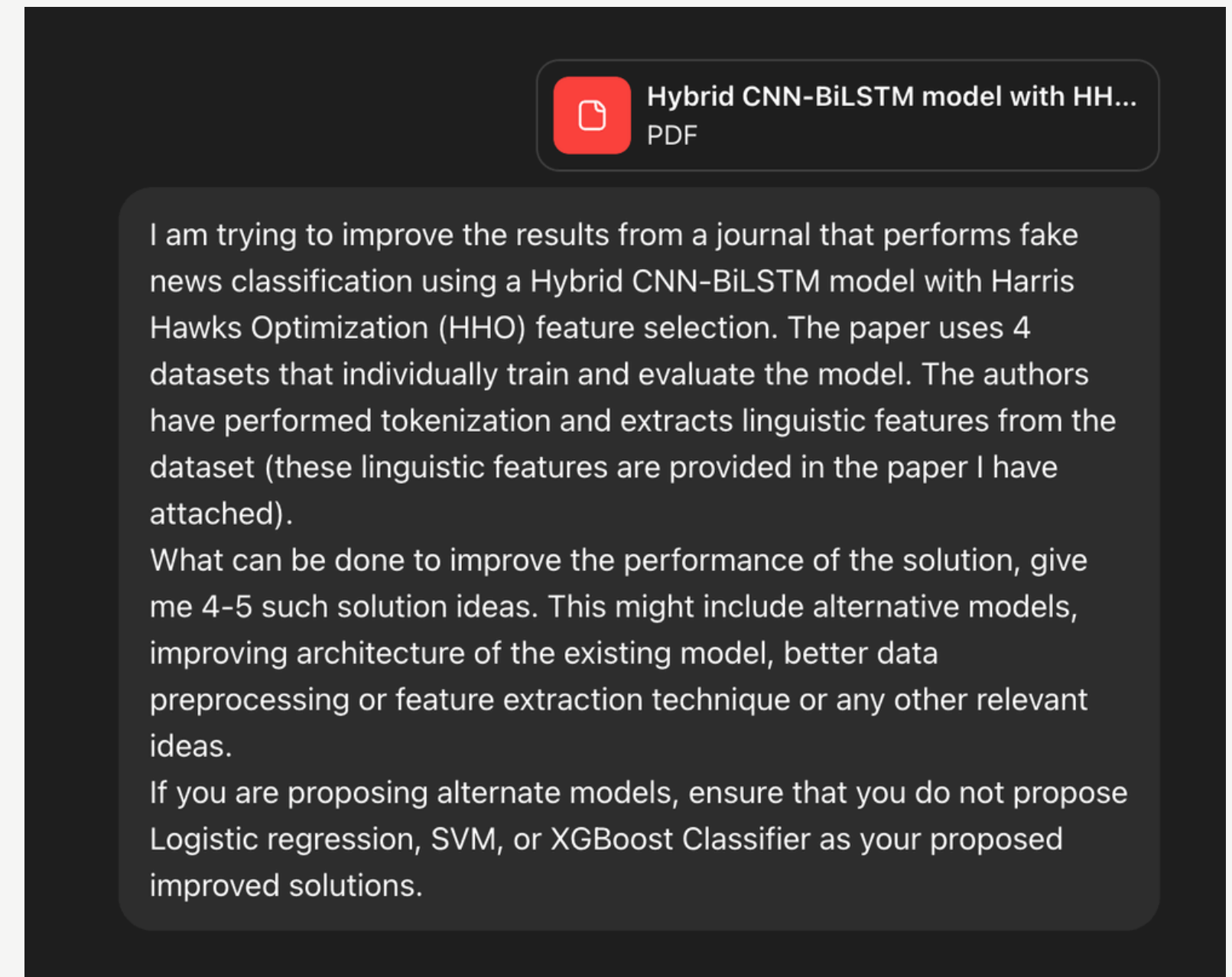
Results - Kaggle Dataset

- The AUC curves for this dataset are more closer together.
- This indicates that the models perform relatively the same.



Discussions & GenAI Implementation

- ChatGPT 5 has been used with extended thinking.
- Prompt introduces the journal article.
- It includes what has already been done.
- Asks for suggested solutions to improve, either through preprocessing, feature extraction, or other models.
- Prompt contains models that should not be considered as they have already been used for this assignment.



Initial Prompt

Solutions Provided

1

Swap the backbone for a document-aware transformer method by fine-tuning it. This would be better for long range dependencies.

2

Upgrade the hybrid architecture & loss by adding a multi-head self-attention pooling. This would surface salient spans of characters.

3

Train for cross-dataset generalization. Implement domain-adversarial training (DANN) to reduce reliance on source-specific quirks.

4

Replace the HHO feature selection with a differentiable feature selection.

Discussions & GenAI Implementation

- Transformer model implemented completely with the help of ChatGPT.
- 125M parameters in the model.
- Model was fine-tuned in 2 stages:
 - Phase 1 with frozen encoders (2 epochs)
 - Phase 2 with encoders unfrozen (5 epochs)
- Model trained on the McIntire dataset due to computational constraints and total training time was around 6 hours.

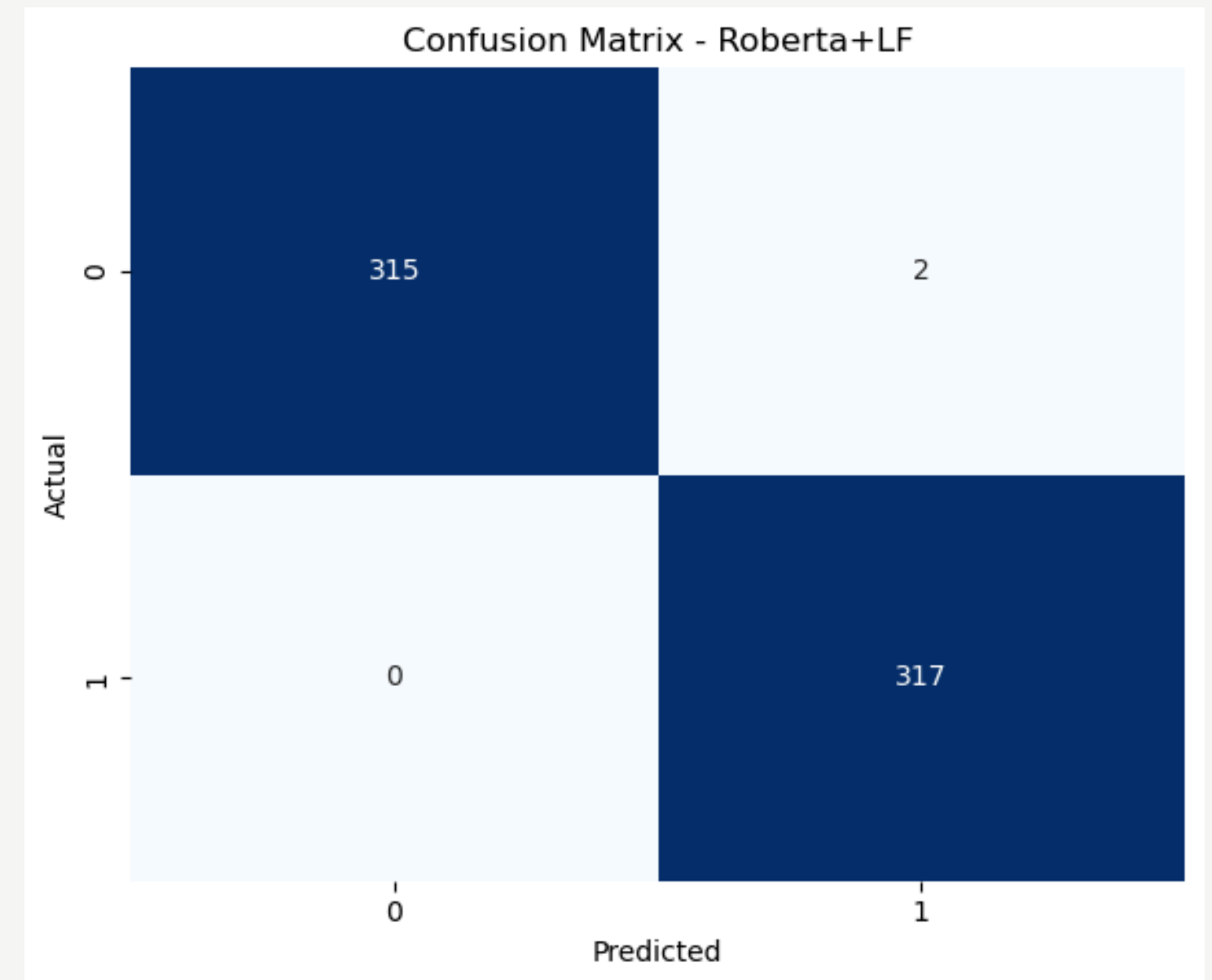
Model: "Roberta_LF_LateFusion"

Layer (type)	Output Shape	Param #	Connected to
lfs (InputLayer)	[(None, 80)]	0	[]
input_ids (InputLayer)	[(None, 510)]	0	[]
attention_mask (InputLayer)	[(None, 510)]	0	[]
layer_normalization (Layer Normalization)	(None, 80)	160	['lfs[0][0]']
tf_roberta_model (TFRobertaModel)	TFBaseModelOutputWithPoolingAndCrossAttentions(last_hidden_state=(None, 510, 768), pooler_output=(None, 768), past_key_values=None, hidden_states=None, attentions=None, cross_attentions=None)	124645632	['input_ids[0][0]', 'attention_mask[0][0]']
dense (Dense)	(None, 128)	10368	['layer_normalization[0][0]']
tf.__operators__.getitem (SlicingOpLambda)	(None, 768)	0	['tf_roberta_model[0][0]']
dropout_38 (Dropout)	(None, 128)	0	['dense[0][0]']
dropout_37 (Dropout)	(None, 768)	0	['tf.__operators__.getitem[0][0]']
dense_1 (Dense)	(None, 128)	16512	['dropout_38[0][0]']
text_proj (Dense)	(None, 768)	590592	['dropout_37[0][0]']
lf_gate (Dense)	(None, 768)	99072	['dense_1[0][0]']
gated_text (Multiply)	(None, 768)	0	['text_proj[0][0]', 'lf_gate[0][0]']
fuse_concat (Concatenate)	(None, 896)	0	['gated_text[0][0]', 'dense_1[0][0]']
dropout_39 (Dropout)	(None, 896)	0	['fuse_concat[0][0]']
dense_2 (Dense)	(None, 256)	229632	['dropout_39[0][0]']
dropout_40 (Dropout)	(None, 256)	0	['dense_2[0][0]']
logits (Dense)	(None, 1)	257	['dropout_40[0][0]']
prob (Activation)	(None, 1)	0	['logits[0][0]']

Total params: 125592225 (479.10 MB)
Trainable params: 125592225 (479.10 MB)
Non-trainable params: 0 (0.00 Byte)

Discussions & GenAI Implementation

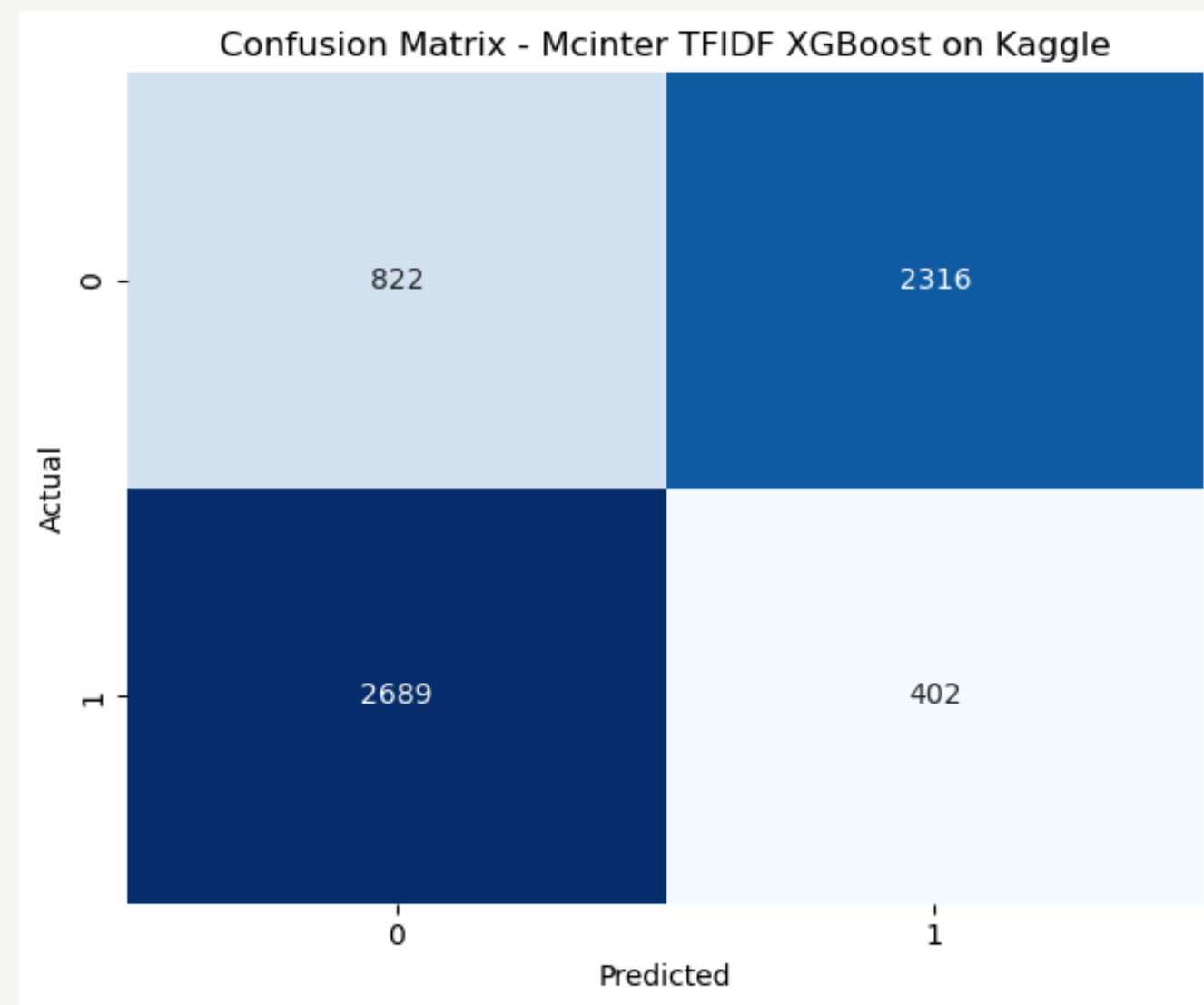
- The model achieved 99.68% test set accuracy, which is far greater than all the other models on the McIntire dataset.
- This is indicative of the benefits of using a transformer-based model to learn those long-range dependencies that we see in textual data.



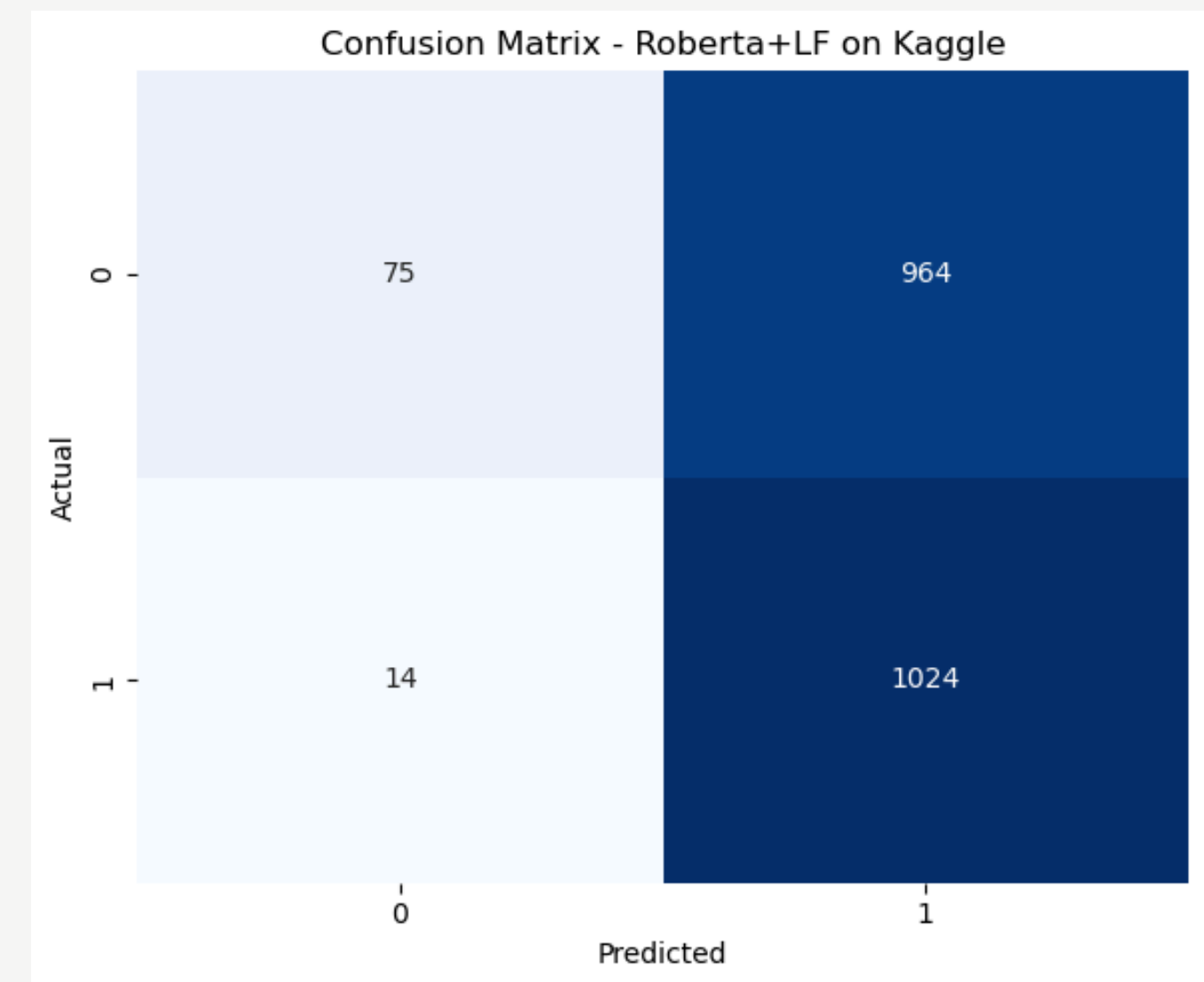
99.68% accuracy

Discussions & GenAI Implementation

To check the OOD generalization of the transformer model, it has been tested on the Kaggle dataset and compared to the McIntire TF-IDF XGBoost model also tested on the kaggle dataset.



19.65% accuracy



52.91% accuracy

Conclusion

- The McIntire and Kaggle datasets were used to reproduce the results of the paper.
- Logistic regression, SVM and XGBoost classifier methods were chosen.
- Hyper-parameter optimization was carried out using a GridSearchCV.
- 2 approaches of those models were analyzed, using Word2Vec embeddings and TF-IDF vectorization. The latter often outperformed.
- The methods used to perform fake news classification are very competitive and often outperform the research article, especially XGBoost Classifier models using a TF-IDF vectorization approach.
- GenAI provided some really good suggestions for improving the results from the paper.
- The transformer method provided by GenAI worked really well overall.
- GenAI did hallucinate, at times providing arguments to the model training that don't exist.

Thank you

———— **For your attention**

Fahad Thakur

11997575
